

Clipforge AI

AI Short-Form Content Creator Studio

Turn Hours of Editing into Minutes of AI.

Project Overview

Field	Details
Student	Haseeb Ahmed
Project	ClipForge AI
Category	AI Short-Form Content Creation Platform
Stage	Functional MVP with Scalable Production Architecture

Executive Summary

ClipForge AI converts long videos into short-form clips with captions, thumbnails, metadata, music, and ZIP-ready exports. It is designed for creators, educators, marketers, podcasters, and small teams that need a faster content repurposing workflow without switching between multiple tools. The platform reduces manual editing time by combining multiple creator tools into one AI-powered workflow.

Problem Statement

Creating consistent short-form content is time-consuming and requires multiple tools. Many creators publish less frequently or stop creating content altogether because they cannot dedicate enough time to editing. Tasks such as clipping videos, adding captions, designing thumbnails, writing metadata, and organizing files separately make the workflow slow, repetitive, and inefficient.

Proposed Solution

The app provides one Creator Studio flow: upload/select video, choose clipping mode, generate shorts, review results, and download a complete package. Current processing includes transcription, segment selection, reframe/crop, styled captions, metadata, thumbnails, music, progress tracking, and ZIP export.

Workflow



Processing uses saved job state and a Redis/RQ worker queue so heavy video work can run outside the FastAPI request process.

Features

1. AI Processing

- Semantic AI highlight detection
- Manual, fixed-duration, and raw footage clipping
- Faster Whisper transcription with transcript caching
- Automatic language detection

2. Smart Editing

- Auto reframe for vertical videos
- Styled captions with custom fonts
- 5 Video Filters (Natural, Punchy, Cinematic, Cool, Black & White)
- English, Roman Urdu, Roman Hindi & Punjabi Captions
- Background music categories

3. AI Content Generation

- AI-generated titles, hooks, descriptions, hashtags, and CTAs
- Three AI thumbnail variations per short
- OpenAI, Hugging Face, and local thumbnail generation

4. Creator Workflow

- Upload, project library, paste-link, and batch processing
- Real-time progress tracking and Resume Job
- Creator Studio with preview and ZIP export

Technology Stack

Layer	Technology Used
Frontend	HTML, CSS, JavaScript static landing page and Creator Studio
Backend	Python, FastAPI, Redis + RQ worker queue
AI / Media	Faster Whisper, OpenAI API, Hugging Face fallback, FFmpeg
Storage	Local Storage, Transcript Cache, Generated Assets, Job Registry
Assets	Caption fonts, music categories, thumbnail overlays, settings files

What Makes ClipForge AI Different?

Unlike traditional editing workflows that require multiple tools, ClipForge AI combines AI clipping, captions, thumbnails, metadata, music, and export-ready packaging into one unified creator workflow. It enables creators to generate complete short-form content from a single upload, saving time, reducing manual effort, and simplifying the entire content creation process.

- One Upload -> Complete Creator Package
- AI-Powered End-to-End Workflow
- Localized Roman Urdu/Hindi/Punjabi Support
- Creator Studio with Organized Export Package.

Future Roadmap

Direct social publishing	Scheduling calendar	Analytics dashboard
Cloud storage and user library	Team workspaces and brand kit	Credits/payment system and API access

Live Demo Includes

Demo Item	Ready
Landing Page	Yes
Creator Studio	Yes
Video Upload	Yes
AI Processing	Yes
Generated Shorts	Yes
Thumbnails	Yes
Metadata	Yes
ZIP Export	Yes

Project Deliverables

Output	Status
Short Videos	MP4
Captions	Styled
Thumbnails	3 per short
Metadata	Title/Hook/CTA
ZIP Export	Ready

Project Summary Status

Area	Current Status	Notes
Frontend	Functional MVP	Landing page, Creator Studio, results dashboard, responsive UI.
Backend	Functional MVP	FastAPI pipeline with local jobs, cache, and queue-ready worker setup.
AI Media	In Progress	Whisper, OpenAI metadata, HF/local thumbnail fallback, FFmpeg rendering.
Production SaaS	Future Roadmap	OAuth, real database, cloud storage, payments, deployment, publishing.

Create More. Edit Less.

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